

Set 1

Below are instructions for each of the exercises in **Set1**.

Go to your **Set1** folder and navigate to the exercise you want to work on.

You can make a workspace for the **Set1** folder in **VS Code** by selecting **Open Folder...** and choosing your **Set1** folder.

This lets you access and manage all your exercise files in one place.

Open the **.c** file you want to run using **VS Code** or any other text editor you are comfortable with (for example, Notepad on Windows).

You can edit your C file directly in any of these editors.

When you finish writing your code, you need to **compile it**.

To do this, open **w64devkit** and navigate to the specific **exercise folder** you are working on within your **Set1** folder.

For detailed steps on how to navigate to the folder and compile your program, refer to the *"Getting started with LEU483"* guide on Canvas.

You do not need to complete all exercises at once. Some can be kept for exam practice.

After completing **Exercise 6**, you are ready to continue with **Set 2**.

Note: Make sure to save your work on the **Z-disc** instead of the **C-disc**.

1 Print text

Open: `ex_printtext/main.c`

Your first task is to open a project and print some text.

Task:

- Make a program that prints out the following text.

```
My first task.  
I fixed it!
```

2 Rectangle

Open: `ex_rectangle/main.c`

Samples: `sample_c2f`, `sample_operators`, `sample_io`, `sample_variables`

This is a program to calculate the area of a rectangle.

Task:

- Ask the user for the width and height of the rectangle. Use this as input to the program.

- Calculate the area of the rectangle. This is the main part of the program.
- Change the format specifier for the output of the program such that the area is printed with 5 decimal places.

Example run of the program (before changing the format specifier):

```
Program to calculate the area of a rectangle
Input the width of the rectangle: 2.1
Input the height of the rectangle: 3.5
The area of the rectangle is : 7.35
```

3 Convert angle unit

Open: ex_convertangle/main.c

Samples: sample_c2f, sample_operators

This is a program to convert an angle between degrees and radians, and vice versa..

Task:

- Find out how to convert between degrees and radians
- Make a program that if 1 is selected, reads an angle in degrees and converts it to radians or if 2 is selected, reads an angle in radians and converts it to degrees.
- Print the result with 4 decimal places.

Example runs of the program:

```
Select                (first run)
1) For deg to rad
2) For rad to deg
> 1
Input angle in deg: 90
90.0000 deg = 1.5708 rad

Select                (another run)
1) For deg to rad
2) For rad to deg
> 2
Input angle in rad: 3.1485
3.1485 rad = 180.3958 deg
```

4 Coin tossing

Open: ex_coin/main.c

Samples: sample_random

Here you are going to simulate tossing (flipping) a coin.

Task:

- Make sure that you understand how to generate 0 or 1 from a random number.
- Let 0 correspond to heads, and 1 to tails. Print the result, such as:

```
Result of coin tossing: head
```

5 Number series

Open: `ex_numberseries/main.c`

Samples: `sample_while_stmt`, `sample_operators`, `sample_incdec`

This is the first exercise in using `while`.

Task:

- Ask the user for an integer number.
- Print all integers divisible with three that are smaller than the selected number. If the user selects 50, the output should be:

```
3 6 9 12 15 18 21 24 27 30 33 36 39 42 45 48
```

6 Dice

Open: `ex_dice/main.c`

Samples: `sample_random`

Here you are going to simulate throwing five dice.

Task:

- Make sure that you can simulate the outcome of throwing one die. That is, you need to generate a random number between 1 and 6.
- Make a loop to generate 5 throws and print the result of each throw. The outcome should be 5 random numbers between 1 and 6. For example it could be:

```
3 5 3 6 1
```

7 Sum-avg

Open: ex_sum_avg/main.c

Samples: sample_if_stmt, sample_while_stmt, sample_incdec

Implement a program to calculate the sum, and average, of some non-negative integers.

Task:

- Read an arbitrary number of non-negative integers as input and calculate the sum and average of these numbers.
- Stop reading numbers by entering -1.

Example run:

```
Calculate sum and average (-1 to quit)
> 1
> 2
> 3
> -1
Sum = 6.000000 Avg = 2.000000
```

Hint: think about what types you use for your variables and how division works for different types.

8 Digits

Open: ex_digits/main.c

Samples: sample_while_stmt, sample_operators

In this exercise you will practice simple "fiddling" with integer numbers.

Task:

- Ask the user for an integer number and print the rightmost digit and the number of digits in this number.

Example run:

```
Input an integer number: 4321
Rightmost digit is 1
Number of digits is 4
```

9 Multiplication

Open: ex_multiplication/main.c

Samples: sample_random, sample_while_stmt

Implement a program for practising multiplication.

Task:

- Ask the user for a multiplication table to practice. This is stored in the variable `table`.
- Generate a new random question from the selected table until the user enters -1.
- For each question, read in one answer from the user and print "Correct" if the answer was correct, or print the correct answer if the user was wrong.
- Count the number of questions and correct answers and print them at the end of the program.

Example output from program:

```
Multiplication practise!
Select table (2-9) to train (-1 to quit): 6
3 * 6 = 18
Correct!
5 * 6 = 30
Correct!
7 * 6 = 44
Sorry, answer was 42!
9 * 6 = 54
Correct!
8 * 6 = -1
Number of questions = 4. Number of correct = 3
```

10 Fixme

Open: ex_fixme/main.c

Samples: sample_io

This code has recurring errors common for novice programmers. Each block of code addresses one or more errors. Can you identify them and correct them?

Task:

- Uncomment one block at the time and correct the errors that you, or the compiler, finds.

11 Systembolaget

Open: ex_systembolaget/main.c

Samples: sample_if_stmt

This is an old exam question. Try to solve this with pen and paper before using the computer.

Task:

Skriv ett program som skulle kunna vara till hjälp i kassan på Systembolaget. Uppgiften är att ta ett födelsedatum och kontrollera om personen är minst 20 år eller ej. Programmet svarar med OK eller !!! Not OK !!!.

Datum matas in med åtta siffror, enligt YYYYMMDD. Dagens datum ska anges på samma sätt, men kan vara "hårdkodat", dvs. vara inskrivet i själva programmet.

Tips: Detta kan lösas utan att veta antalet dagar i en månad (eller ett år).

Ett exempel med dagens datum 20210826 (man måste vara född senast 20010826 för att få "köpa ut", dvs. för att programmet ska ge OK):

```
Personal number: 20010827
!!! Not OK !!!
```